

MA 603: Advanced Partial Differential Equations

Course Code	MA 603
Course Name	Advanced Partial Differential Equations
L-T-P-C	3-0-0-3
Prerequisites	MA 522 Partial Differential Equation
Intended for	M.Sc./B.Tech/M.S./M.Tech/Ph.D.
Distribution	Elective
Approval	30th BoA (Jun 2019); 22nd Senate (Jun 2019)

Course Contents

- **Green's Function** [4 Lectures]
Green's functions, Green's functions and applications for Laplace, Poisson and Helmholtz equations Green's functions and applications for the heat equation Green's functions and applications for the wave equation.
- **Non Linear First Order PDE** [6 Lectures]
Complete Integrals and New Solutions from Envelopes, Local Solution and Application. Equations that convert into linear PDE; some exactly solvable cases; Burgers' equation; dimensional analysis and similarity; travelling waves; nonlinear diffusion and dispersion. Introduction of Hamilton Jacobi Equations, Calculus of Variations, Hamilton's O.D.E., Boundary Conditions, Local Solutions and Applications.
- **Module III** [3 Lectures]
Asymptotics, Singular perturbations, Turing Instability for Reaction Diffusion System, Laplace's Method, Homogenization, Power Series, Non Characteristic Surfaces, Real analytic Functions, Cauchy Kovalevskaya Theorem.
- **Sobolev Spaces** [12 Lectures]
Introduction to Hilbert Spaces of Functions, Holder Spaces, Sobolev spaces; Definitions and Elementary Properties. Weak solution, Uniqueness and Properties of Weak derivatives. Definition & Properties of Sobolev Spaces, Inequalities, Compactness. Extensions, Sobolev
- **Second Order Elliptic Equations** [8 Lectures]
Weak Solutions of Elliptic Equation, Existence of weak solutions, Regularity, Maximum Principles, Eigen values & Eigen Functions of symmetric elliptic operators.
- **Reaction Diffusion System** [9 Lectures]
Weak Solution of Diffusion Equation, Green's Function of Diffusion Equation, Formulation of Reaction Diffusion models and extensions to Include Chemotaxis terms; Application of Reaction Diffusion Systems to Population Dynamics, Pattern and Wave Phenomenon in the Life Sciences, Semi-arid vegetation and wound healing as Prototype Examples

Text Books

1. Evans, Lawrence C., **Partial Differential Equations**, Graduate Studies in Mathematics, vol. 19. Providence, RI: American Mathematical Society, 2010.
2. Brezis, H. And H. Brezis., **Functional Analysis, Sobolev Spaces and Partial Differential Equations**, Springer, 2011.

References

1. Debnath, Lokenath, **Nonlinear partial differential equations for scientists and engineers**, Springer Science & Business Media, 2011.
2. DiBenedetto, Emmanuele. **Partial Differential Equations**, Birkhauser, 1995.
3. Garabedian, Paul, **Partial Differential Equations**, Providence, RI: AMS Chelsea, 1998.
4. E. Kreyszig, **Advanced Engineering Mathematics**, Wiley, 2011.